Dionysios Adamopoulos

Richard-Wagner-Straße 63, 66111, Saarbrücken, Germany - dionadam2013@gmail.com

Education

- **Max Planck Institute for Software Systems** (MPI-SWS), Saarbrücken, Germany
  PhD, Computer Science *(April 2026 - Today)*
  - Advisor: Dr. Christina Giannoula
- **National Technical University of Athens** (NTUA), Athens, Greece
  Integrated Master's Degree, Electrical and Computer Engineering *(September 2020 - February 2026)*
  - GPA: **9.18/10** – "Excellent"
  - Credits: **300 ECTS**
  - Thesis: Analysis and Optimization of Sparse Convolution on GPUs (grade: **10/10**)

Research Experience

- **Research Intern, Max Planck Institute for Software Systems** (MPI-SWS), Saarbrücken, Germany *(November 2025 - March 2026)*
  - Advisor**:** Dr. Christina Giannoula
  - Conducting research at the intersection of Computer Architecture, Systems and Machine Learning
- **Research Intern, Huawei Munich Research Center**, Munich, Germany *(November 2023 - January 2024)*
  - Internship at the Applied Network Technology Laboratory
  - Studying IoT routing protocols (e.g., AoDV, OLSRv2) and implementing an experimental routing protocol in the OMNet++ discrete event simulator and proposed algorithmic improvements

Scholarships, Competitions & Awards

- **Mathematics Competitions**
  - **Third Prize Award**, International Mathematical Olympiad (**IMO**), 2020
  - **Third Prize Award**, Balkan Mathematical Olympiad (**BMO**), 2020
  - **Gold Medal (1st Country Rank)**, Greek National Mathematical Olympiad, 2020
  - **Third Prize Award**, Mediterranean Youth Mathematical Championship (**MYMC**) 2019
  - **Second Prize Award**, Greek National Mathematical Olympiad, 2019
  - **Third Prize Award**, Greek National Mathematical Olympiad, 2018
- **Informatics Competitions**
  - IEEEXtreme 2024 (Competitive Programming Team Competition)**:**
    - Country Rank: **2/112**, Global Rank: **60/5801**
  - Greek National Competition of Informatics (2018):
    - 12st place in national rank and admission to the competitive programming preparation camp
- **PAD Scholarship (Pädagogischer Austauschdienst) – German Ministry of Education** (2017)
  - Selected as one of the **top 4** students from Greece for an academic exchange program in Germany

Publications

- "Spira: Exploiting Voxel Data Structural Properties for Efficient Sparse Convolution in Point Cloud Networks", **D. Adamopoulos,** A. Poulopoulou, G. Goumas, C. Giannoula - **MLSys 2026 (Link)**
- "Sparse Computations in Deep Learning Inference**"**, I. Tasou, …, **D. Adamopoulos**, et al. - ArXiv Preprint **(Link)**

Notable Projects

- Designed and developed a **compiler** for a Pascal-like language using OCaml **(GitHub)**
- **Open-Source Implementation** of "Spira: Exploiting Voxel Data Structural Properties for Efficient Sparse Convolution in Point Cloud Networks" - **MLSys 2026 (GitHub)**

Technical Skills

- **Langages & Frameworks:** C, C++, CUDA, Python, Javascript, OCaml, Pytorch, LLVM
- **Parallel Programming:** MPI, OpenMP, GPU Programming
- **Tools & Simulators:** Nvidia Nsight Profiling Tools (Systems Analysis), PIN (Dynamic Binary Instrumentation), Git, OMNet++ (Discrete Event Simulator)

Languages

- **English:** Proficient, Certificate of Proficiency in English, **University of Michigan**
- **German:** Fluent, Goethe-Zertifikat B2, **Goethe-Institut**

Volunteering

- **Teaching Assistant:** Assistant at the undergraduate courses "Programming Techniques" and "Operating Systems"
- **Volunteer Grader:** Grading Assistant for national mathematics competitions (Hellenic Mathematical Society)